Supplementary material for "RePressDB: A Dependable Self-Adaptive Storage Architecture for Nonstationary Time-Series Workloads"

Zhichao Zhang[a,*], Xiaolu Peng[b,1], Daoqi Sun[a,1]

[a]*School of Science, Minzu University of China, No. 27 Zhongguancun South Avenue, Beijing, 100081, China*
[b]*School of Navigation, Dalian Maritime University, No. 1 Linghai Road, Ganjingzi District, Dalian 116026, China*

1. Purpose, numerical discipline, and experimental records

This supplement records the evaluation details, configurations, and supporting measurements used in the study. Result tables are generated from archived CSV or JSON evidence, and derived values are propagated through shared numerical macros. The two archived experiment packages are:

Table 1: Archived experiment packages.

Evidence package	SHA-256
J2/J3/J4 experiment package (v1.5.1)	`19aad963a625e14c2298bb783bc4885d ff062ae17427200f0f1cfbcb7713ed39`
Paired end-to-end package (v1.7.0)	`ff773467a5de296e07dcbaac904b2d26 c466a815ca7b16bdcddbff527a2c4088`

The v1.5.1 package freezes independently seeded measured-cost replay, unified ablation, randomized termination/corruption, and calibration studies. The v1.7.0 package freezes the paired uncached-versus-active-view end-

*Corresponding author: Zhichao Zhang (zhichao.zhang2024@outlook.com).
Email addresses: zhichao.zhang2024@outlook.com (Zhichao Zhang), lilypxl1656179423@outlook.com (Xiaolu Peng), sundao0217@gmail.com (Daoqi Sun)
[1]Xiaolu Peng and Daoqi Sun contributed equally to this work and are co-second authors.

to-end campaign. Neither package is used to retune RePressDB or Shape-EWMA-BE.

2. Controller and baseline configurations

2.1. RePressDB

Table 2: Controller settings.

Setting	Value
Long/short windows	300 / 100 local queries
Minimum observations	200 local queries
Decision interval	100 local queries
Confirmation rounds	3
Cooldown	300 local queries
Projection horizon	1,200 local queries
Minimum relative gain	8%
Migration safety factor	1.25
Global budget interval	1,000 global queries
Budget bytes	Largest measured directed rewrite in the corresponding cost table

2.2. Shape-EWMA-BE and Page-Hinkley-BE

Shape-EWMA-BE was selected only on the PEMS04 development data. The search contained 80 configurations; 44 passed the pre-registered brief-burst and stable-scan safety gates. The selected configuration was within 0.5% of the best action score and minimized migrations and rewritten bytes within that equivalence set.

Table 3: Baseline configurations.

Baseline	Configuration and runtime information
Shape-EWMA-BE	EWMA $\alpha = 0.01$, relative deadband 0.05, horizon 300, decision interval 100, minimum observations 200, and measured directed break-even. It observes only past query classes.
Page-Hinkley-BE	Page-Hinkley $\delta = 0.005$, threshold 10, window cap 300, deadband 0.05, horizon 600, and the same measured directed break-even. It observes only past query classes.
AF-HC-D	Access-frequency hot/cold detector with a deadband and the same write budget; decisions use access frequency instead of the eight-class query-shape monitor.
Initial-best	Deployable static representation chosen for the fixed initial scan-heavy mixture.
Best-static	Nondeployable hindsight reference selected from the complete trace with a free initial choice.
Migration-aware oracle	Nondeployable future-aware reference that sees phase boundaries and future duration.

3. Multi-seed measured-cost replay

The statistical unit is one independently generated trace realization fixed before policy execution. The campaign contains three trace families, two sizes, and three pre-registered seeds, giving 18 realization-level cells. Ratios use equal-weight geometric means. Query and directed migration costs use file-backed medians from the archived calibration tables; system-level timing is measured separately in the paired file-backed campaign.

Table 4: Multi-seed replay results.

Trace	Size	RePress/Shape	RePress/initial	RePress/PH-BE	RePress mig.	Shape mig.
Gradual-periodic	8K	0.9453	0.9453	0.9590	44	0
Gradual-periodic	16K	0.9586	0.9067	0.9956	55	40
Mean-reverting	8K	0.9806	0.9806	1.0008	6	0
Mean-reverting	16K	0.9541	0.9355	0.9624	41	170
Noisy bursts	8K	0.9795	0.9795	0.9986	6	0
Noisy bursts	16K	0.9527	0.9472	0.9648	33	151
All cells	N/A	0.9617	0.9488	0.9800	185	361

Across all 18 cells, RePressDB/Shape-EWMA-BE is 0.9617 (bootstrap 95% CI 0.9551-0.9689), and RePressDB is lower in 18/18 cells. RePressDB/initial-best is 0.9488, RePressDB/Page-Hinkley-BE is 0.9800 with 15/18 lower cells, and RePressDB/free best-static is 1.0564. Best-static has complete-trace knowledge and a free initial choice, so this comparison is interpreted as an optimistic offline reference.

Table 5: Aggregate migration-risk results.

Policy	Migrations	Unamortized rate	Stale rate
AF-HC-D	997	66.90%	43.70%
Page-Hinkley-BE	708	71.89%	30.96%
Shape-EWMA-BE	361	70.91%	22.19%
RePressDB	185	8.65%	12.43%

4. Unified component ablation

The six variants use the same plans, profiles, migration matrices, and write-budget accounting. The ablation is interpreted through the joint behavior of replay cost, migration count, and migration-risk measures.

Table 6: Ablation results.

Variant	Cost ratio	95% CI	Migrations	Median unamortized	Median stale
Full RePressDB	1.0000	[1.0000, 1.0000]	66	3.33%	2.78%
No break-even	0.9667	[0.9428, 0.9886]	130	16.23%	16.23%
No confirmation	0.9857	[0.9600, 1.0102]	229	10.51%	0.00%
No cooldown	1.0000	[1.0000, 1.0000]	66	3.33%	2.78%
No write budget	0.9994	[0.9988, 0.9999]	66	3.33%	2.78%
Short-window only	0.9941	[0.9803, 1.0073]	126	8.33%	2.63%

Removing break-even changes replay cost to 0.9667 of full RePressDB while migrations increase from 66 to 130 and the median unamortized rate increases from 3.33% to 16.23%. Removing confirmation raises migrations to 229. In these six cells, cooldown and the write-budget constraint are activated infrequently enough that aggregate cost and migration count remain close to full RePressDB; their role is to bound temporal reversals and aggregate write activity.

4.1. Clean-stage responsiveness holdout

The clean-stage holdouts isolate response under abrupt, persistent changes. Shape-EWMA-BE reacts earlier to these transitions, while RePressDB performs fewer migrations and rewrites fewer bytes across the same PEMS08 and BDG2 cells. The figure reports both sides of this responsiveness/selectivity trade-off.

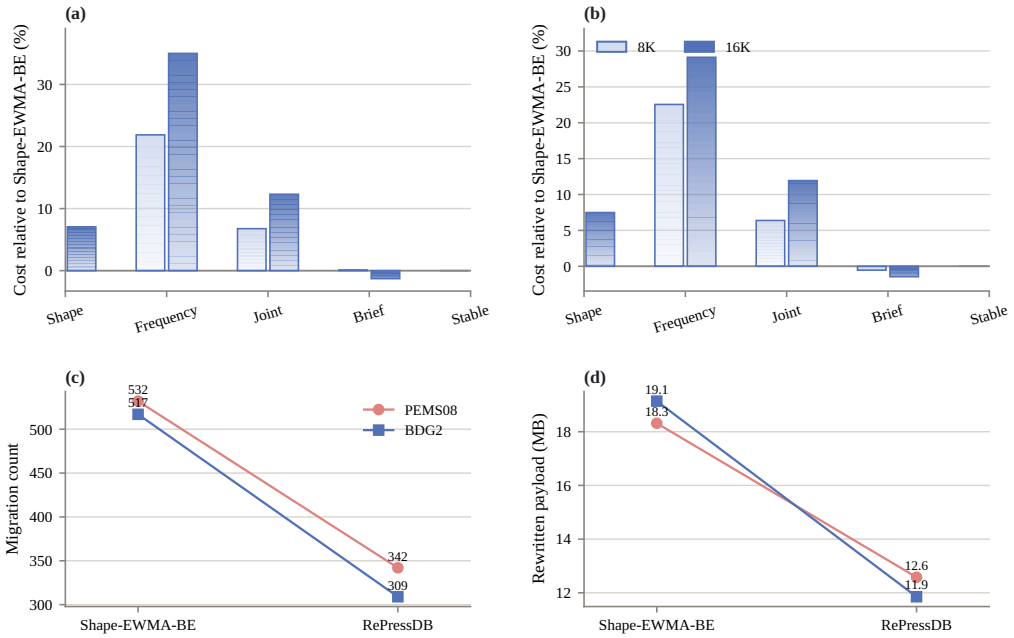


Figure 1: Clean-stage holdout results.

5. Paired end-to-end campaign

5.1. Process structure and pairing

The v1.7.0 launcher retains only Python standard-library modules. It starts one heavy worker, waits for it to exit, and starts the next worker only after the preceding process has released its NumPy, Pandas, codec, SQLite, and result state. The campaign contains 54 paired units and 108 isolated workers. Each pair shares trace, size, repeat, policy, calibration table, initial store snapshot, and query realization. Uncached-first and active-view-first orders are balanced 27/27. The total measured query count is 16,908,066, and every query is checked against an independent `float32` truth array.

Each worker performs one verification query per active chunk before the timed region. This loads the current representation and metadata and matches the `warm_slicedmeta` calibration regime. The prewarm cost is reported separately. A post-migration active-view reload remains inside the measured interval.

Table 7: Active-view execution effect.

Policy	Service ratio	95% CI	Dispatch ratio	p99 ratio	Cached faster	Pairs
Initial-best	0.4718	[0.3778, 0.6084]	0.0329	0.4548	16/18	18
Shape-EWMA-BE	0.3560	[0.3400, 0.3734]	0.0198	0.3333	18/18	18
RePressDB	0.2638	[0.2168, 0.3316]	0.0123	0.3761	17/18	18

5.2. Read-path ablation

Table 8: Read-path reconstruction counts.

Counter	Uncached	Active view
Catalog-row loads	16,909,113	246
`StoredChunk` builds	8,454,711	246
Block-metadata parses	2,358,417	246

The uncached path yields a RePressDB/Shape service-time ratio of 1.4142. Its accumulated time decomposition is 117.10 versus 106.98 s for codec execution, 186.28 versus 105.31 s for catalog/dispatch, 5.22 versus 6.91 s for controller work, and 2.23 versus 1.14 s for migration work. Repeated catalog and metadata reconstruction account for most of the additional read-path work.

All 54 uncached/active-view pairs have identical migration signatures: event index, chunk, source codec, target codec, and rewrite bytes. Active-view workers incur 93 RePressDB, 30 Shape, and zero initial-best measured cache misses, exactly matching the committed migration counts. The paired design therefore isolates the read-path mechanism while keeping adaptation decisions fixed.

5.3. Policy comparison with active-view execution

Table 9: Policy ratios with active-view execution.

Comparison	Service ratio	95% CI	Median	Wins
RePressDB / Shape-EWMA-BE	0.9575	[0.7605, 1.2027]	1.0009	9/18
RePressDB / initial-best	0.9896	[0.8379, 1.1302]	1.1014	4/18

The aggregate service-time ratios remain close across the adaptive policies. RePressDB/Shape p99 is 1.1256 (95% CI: 0.8470–1.5159), and RePressDB/initial-best p99 is 1.3803 (95% CI: 1.0653–1.7535). Three paired units contain large codec-time observations; they remain in every aggregate and confidence interval.

6. Recovery campaign and state coverage

Table 10: Randomized recovery campaign.

Trial class or pre-termination state	Count
Abrupt termination: `idle`	110
Abrupt termination: `intent`	84
Abrupt termination: `encoded`	23
Abrupt termination: `validated`	31
Abrupt termination: `published`	28
Abrupt termination: `committed_pending_cleanup`	24
Post-commit payload truncation	75
Post-commit metadata corruption	50
Orphan staging/final files	75
Randomized campaign total	500
Earlier controlled protocol-boundary trials	600
Combined tested trials	1,100

All 500 randomized trials finish with a valid idle catalog, exact full-array equality, 20 correct spot queries, no temporary file, no invalid active chunk, no non-idle row, and exactly one active payload. Together with the earlier 600 controlled boundary trials, all 1,100 completed trials satisfy the recorded recovery and query-correctness checks. The evaluated fault model covers process interruption and selected corruption conditions. Hardware-level power interruption and storage write reordering belong to a stronger durability campaign.

7. Calibration overhead and robustness

Fresh calibration covers 64 chunks, five codecs, eight query classes, 2,560 query-cost records, 320 codec records, and 1,600 source-to-target entries. The last count includes five no-op diagonal entries per chunk; there are 20 nontrivial directed migrations per chunk.

Table 11: Calibration overhead.

Metric	Value
Wall time	26.10 s
CPU user/system	8.31 / 5.39 s
Process reads	923.23 MB
Process writes	113.15 MB
Raw chunk bytes	3.15 MB
Candidate storage	12.40 MB
Candidate/raw ratio	3.94×

Table 12: Sampled-profile evaluation.

Sample fraction	RePressDB true-cost ratio	Shape true-cost ratio	Median migration Jaccard (RePressDB)
10%	1.0859	1.0675	0.000
25%	1.0835	1.0433	0.073
50%	1.0938	1.0265	0.131

A population-median profile built from a subset of chunks changes chunk-specific migration decisions. Independent bounded perturbations of the existing per-chunk cost entries have much smaller effects: at ±20%, RePressDB's geometric-mean true-cost ratio is 1.0004 and its p95 ratio is 1.0069. These two sensitivity studies separate loss of chunk-specific structure from bounded local measurement error.

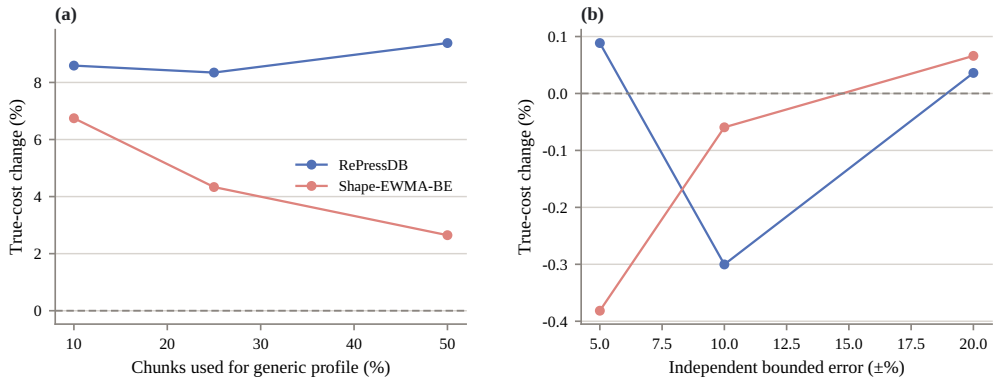


Figure 2: Calibration sensitivity.

8. Physical corpus, logical scale, and interference

The full cleaned-BDG2 physical campaign contains 564 eligible series, 720 complete 8K chunks, and 3,283 2K catalog chunks. It completes one million verified file queries and 1,000 migrations. This physical campaign is distinct from the shared-template logical-scale experiment.

Table 13: Logical catalog scaling.

Logical scale	Chunks	Tenants	Catalog B/chunk	Clean start (s)	One-dirty-row start (s)
1 GiB	32,768	46	301.5	0.283	0.317
5 GiB	163,840	228	298.6	2.656	2.958
10 GiB	327,680	456	300.5	5.526	6.190

The 10-GiB label denotes a logical catalog and routing view over 327,680 chunks backed by shared physical templates. Across nine logical-scale/thread strata, the counterbalanced background-migration campaign gives an aggregate foreground-throughput ratio of 0.9764 and a p99 amplification of 1.0569. Each background cell contains a bounded burst of 50 copy-on-write migrations.

9. Captured environment and reproducibility details

Table 14: Experimental environment.

Item	Captured value
Operating system/filesystem	Microsoft Windows 11 Home, build 26200; NTFS
Processor	Intel Core i7-14700HX; 20 physical cores / 28 logical processors
Memory	16,873,545,728 bytes (16.87 GB; about 15.7 GiB)
Storage	YMTC YMSS2ED08D25MC; 1.024 TB; NVMe; firmware LN3603D1
Python	3.10.20
NumPy / Pandas	2.2.6 / 2.3.3
SQLite	3.53.1; WAL; `synchronous=FULL`
LZ4 / Zstandard	4.4.5 / 0.25.0; whole Zstandard level 3
Data type	Little-endian `float32`
Timer	`time.perf_counter_ns`
End-to-end process model	Standard-library launcher; one isolated heavy process per worker; no additional memory-pressure allocation

A same-host hardware capture records the processor and project-drive identifiers reported above. Absolute latency and throughput remain machine-specific. The principal conclusions therefore rely on balanced paired ratios, confidence intervals, medians, and win counts.

10. Artifact layout and interpretation scope

The public repository at Jasper-Cheung/RePressDB-JSS-Artifact contains source code, environment records, selection manifests, archived summaries, and generation scripts. Versioned archives distributed through the repository's versioned releases add all 108 worker summaries, raw latency arrays, recovery and calibration records, and SHA-256 manifests. Public source datasets remain available from their cited repositories.

Measured-cost replay and wall-clock execution are reported separately throughout the study. End-to-end results use the warm-metadata active-view path and balanced paired units. The 10-GiB scale experiment concerns catalog and routing behavior over shared immutable templates. Recovery evidence covers the specified process-interruption and corruption model, with stronger hardware-level durability conditions treated as a separate validation layer. The representation space contains five lossless layouts, and the evaluation uses one file-backed host with real sensor values and controlled or semi-synthetic query flows.